

Thermal Stress Stability Study

Sterility, bacterial endotoxin, and potency of five compounded sterile preparations after a continuous 13-day 17-hour high-temperature excursion.

137.1°F = 58.4 °C peak

MAXIMUM RECORDED TEMPERATURE • LOGGER S/N
EFE261104293

RECORDED PROFILE °F — 24 JAN → 06
FEB 2026



01 Result at a glance

Every lot subjected to the excursion met all acceptance criteria on subsequent laboratory testing.



All five compounded preparations passed post-excursion release testing.

After continuous exposure peaking at **58.4 °C (137.1 °F)** over **13 d 17 h**, all lots returned **no evidence of microbial growth** (USP <71> sterility) and endotoxin levels **far below** their acceptance limits (USP <85>). The Sermorelin lot additionally assayed at **102.3 % of label potency** by HPLC. Testing performed by Pharmetric Laboratory (accreditation 5386.01).

Minimum

28.8°F

-1.8 °C

Average

66.9°F

19.4 °C

MKT

77.2°F

25.1 °C

Maximum

137.1°F

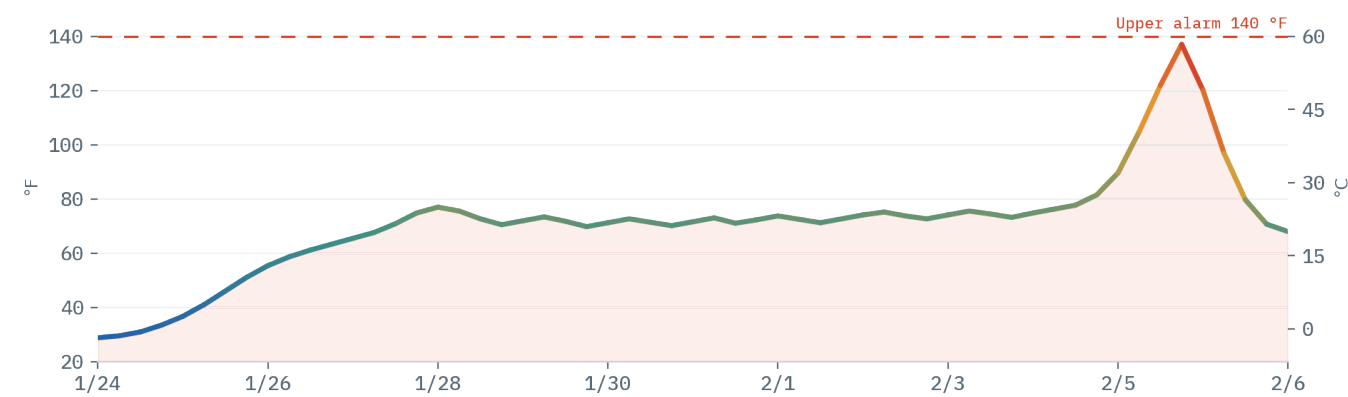
58.4 °C

02 Exposure conditions

A calibrated RC-5 data logger recorded temperature every 15 minutes throughout the study. The trace below is a faithful redraw of the instrument's own plotted profile; summary values are the logger's recorded statistics.

Temperature vs. alarm thresholds

Dual axis — °F (left) and °C (right). Upper alarm (H1) 140 °F / 60 °C; lower alarm (L1) -22 °F / -30 °C. No alarm state was triggered.



Logger & configuration

Model
RC-5
Serial no.
EFE261104293
Probe
Temperature
Firmware
V2.1
Logging interval
15 min
Total readings
1,317
Trip code
0000001

Study window

First reading
2026-01-24 06:01
Last reading
2026-02-06 23:01
Duration
13 d 17 h
Upper alarm (H1)
60.0 °C / 140 °F — Ok
Lower alarm (L1)
-30.0 °C / -22 °F — Ok
Peak reached
58.4 °C / 137.1 °F
File created
2026-06-03 16:42

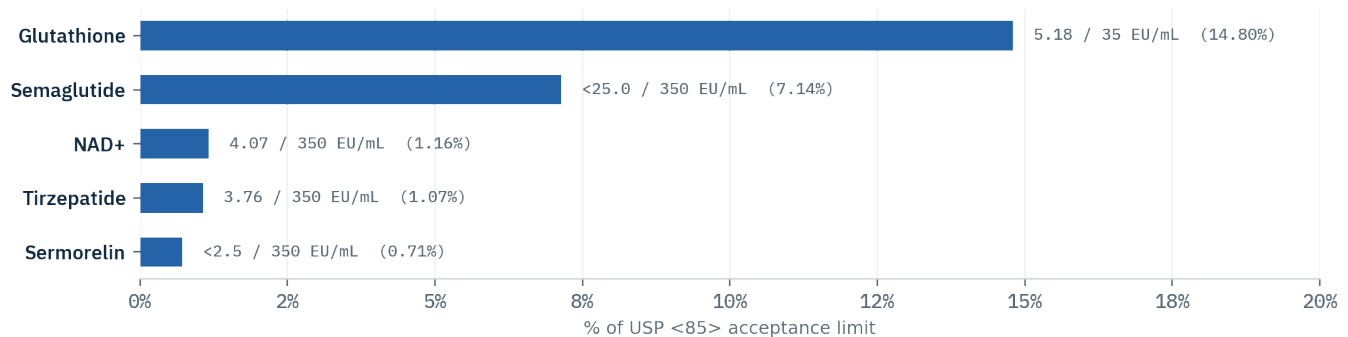
03 Post-excursion test results

Certificates of Analysis issued by Pharmetric Laboratory, 7265 Ulmerton Road, Largo, FL. Sterility by ATP bioluminescence (USP <71> compliant); endotoxin by USP <85>.

Preparation	Lot / Sample	Sterility <71>	Endotoxin <85>	Potency	Released
Tirzepatide / Glycine ₃₀ / 20 mg/mL · 2 mL vial	LG396001032S429010	No growth	3.76 / 350 EU/mL	—	05 May 26
Semaglutide / Glycine ₅ / 20 mg/mL · 2 mL vial	LG396001001S429007	No growth	<25.0 / 350 EU/mL	—	08 May 26
NAD ⁺ Injectable 100 mg/mL · 10 mL vial	04162026@10S429006	No growth	4.07 / 350 EU/mL	—	06 May 26
Glutathione 200 mg/mL · 10 mL vial	Lg396000890S429004	No growth	5.18 / 35 EU/mL	—	05 May 26
Sermorelin 2 mg/mL · 5 mL vial	04162026@12S423999	No growth	<2.50 / 350 EU/mL	102.3 %	22 Apr 26

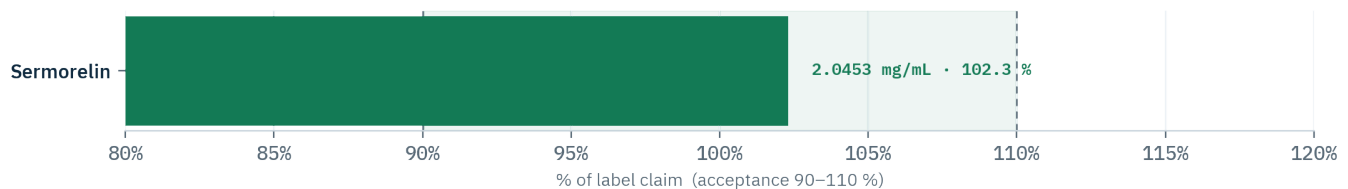
Endotoxin headroom — result as % of allowable limit

Each bar shows the measured endotoxin as a fraction of that product's USP <85> acceptance limit. Shorter is better; all lots sit well inside the limit (100 %).



Sermorelin potency — HPLC assay

Measured 2.0453 mg/mL against a 2 mg/mL label claim. Acceptance range 90.0–110.0 %.



Scope note: Sterility and bacterial endotoxin were performed on all five lots. A potency (HPLC) assay was performed on the Sermorelin lot; the remaining lots were qualified for sterility and endotoxin only. Results relate to the item(s) tested and to the samples as received by the laboratory.

04 Shipment & chain of custody

UPS Ground Proof-of-Delivery records accompanying the study samples.

Shipment 1 — San Diego, CA

Tracking
1ZV543K10305361985
Service
UPS Ground
Shipped / billed
04/16/2026
Delivered
04/22/2026 1:17 PM
Location
Security Gate

Shipment 2 — Tampa, FL

Tracking
1ZV543K10326600883
Service
UPS Ground
Shipped / billed
04/21/2026
Delivered
04/29/2026 11:42 AM
Received by
ALIANA — Inside

05 Conclusion

Products retained sterility, endotoxin control, and potency after sustained heat.

Across a continuous 13-day 17-hour excursion that peaked at 58.4 °C (137.1 °F) — approaching the logger's 60 °C / 140 °F upper alarm — every compounded lot met its release specifications on subsequent testing:

- 5 of 5 lots — no evidence of microbial growth (USP <71> sterility)
- 5 of 5 lots — bacterial endotoxin below acceptance limit (USP <85>); measured values ranged from <2.50 to 5.18 EU/mL
- Sermorelin — 102.3 % of label potency by HPLC, within the 90–110 % range
- No alarm threshold was breached during the recorded exposure window



Tradeline SH Inc. d/b/a Logos Pharmacy · 7543 W Waters Ave, Tampa, FL 33615
Laboratory testing by Pharmetric Laboratory, LLC (accreditation 5386.01). Temperatures shown in °C with °F equivalents (°F = °C × 9/5 + 32).
This report summarizes source documents provided by Logos Pharmacy and is intended for internal quality and informational use. Values transcribed from the underlying CoAs and logger report.

Proof of Delivery

Dear Customer,

This notice serves as proof of delivery for the shipment listed below.

Tracking Number

1ZV543K10305361985

Service

UPS Ground

Delivered On

04/22/2026 1:17 P.M.

Delivered To

SAN DIEGO, CA, US

Shipped / Billed On

04/16/2026

Delivery Location

Security Gate



Sincerely,

UPS

Tracking results provided by UPS as of 05/15/2026 2:09 P.M. EST

Proof of Delivery

Dear Customer,

This notice serves as proof of delivery for the shipment listed below.

Tracking Number

1ZV543K10326600883

Service

UPS Ground

Delivered On

04/29/2026 11:42 A.M.

Delivered To

TAMPA, FL, US

Shipped / Billed On

04/21/2026

Received By

ALIANA

Delivery Location

Inside Delivery

Sincerely,

UPS

Tracking results provided by UPS as of 05/15/2026 2:10 P.M. EST

Device Information

Model:	RC-5	Probe Type:	Temperature
S/N:	EFE261104293	Firmware Version:	V2.1
Total Space:	32000		

Trip Description

Trip Code:	0000001
Trip Description:	Temperature recording.

Config. info

Button Stop:	Enable	Logging Interval:	15m
Button Tone:	N/A	Alarm Logging Interval Shorten (1 min):	N/A
Alarm Tone:	N/A	Storage Mode:	N/A

Alarm Threshold

H1: Above: 60.0°C
Ideal Zone:
L1: Below: -30.0°C

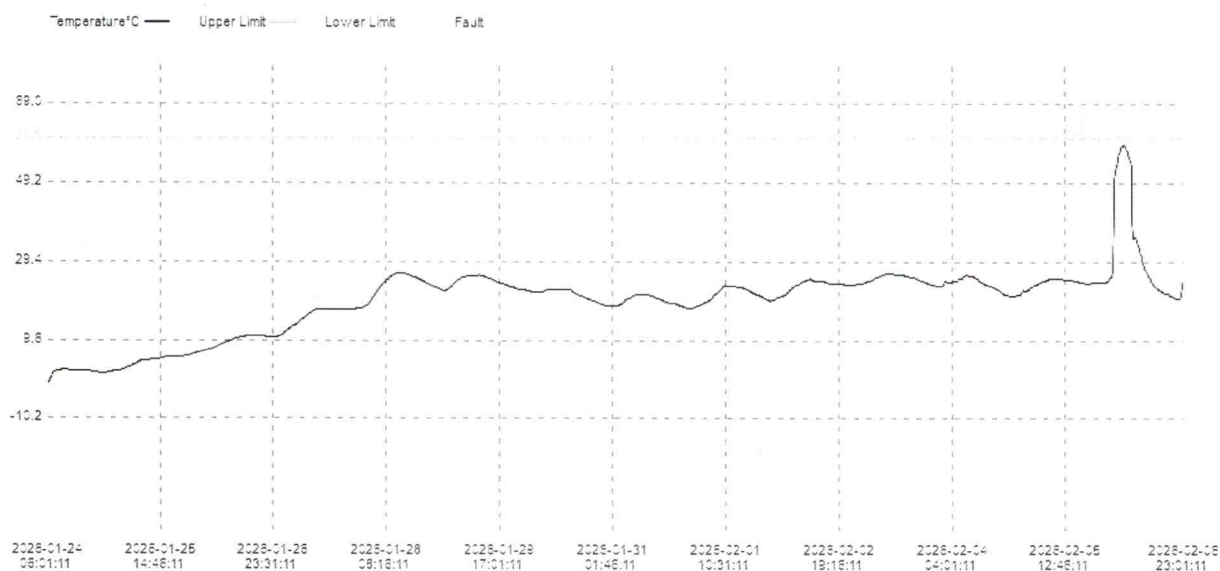
Alarm

Ok

Ok

Summary

Maximum:	58.4°C	First Reading:	2026-01-24 06:01:11
Minimum:	-1.8°C	Last Reading:	2026-02-06 23:01:11
Average:	19.4°C	Logging Duration:	13d 17h
MKT:	25.1°C	Total Memory:	1317
First Alarm(Te):	N/A		





PHARMETRIC
LABORATORY

CERTIFICATE OF ANALYSIS (CoA)

Pharmetric Laboratory
265 Ulmerton Road
Largo, FL, 33771

Issue #:	0	SAMPLE #:	S429010
CoA RELEASE DATE:	05 May 2026		

CLIENT:	Logos Pharmacy	CONTACT:	Carlos Martinez		
CLIENT ADDRESS:	7543 W Waters Ave, Tampa, FL, 33615				
DESCRIPTION:	TIRZEPATIDE 30MG/ML/GLYCINE 20MG/ML (2ML)	LOT #:	LG396001032		
STRENGTH:	30/20 MG/ML	CONTAINER:	2.0 mL Vial		
STORAGE CONDITIONS:	Refrigerated (2°C to 8°C)	BATCH SIZE:	250	EXP.:	Not Provided
DATE(S) RECEIVED:	30 Apr 2026	QUANTITY RECEIVED:	12		

TEST DATE	TEST	QTY	ACCEPTANCE CRITERIA	RESULTS	COMPLIANCE STATEMENT
05 May 2026	Sterility-ATP Bioluminescence Test	10	No Evidence of growth	No Evidence of growth	PASS
02 May 2026	USP <85> Endotoxin Test	1	350 EU/mL	3.76 EU/mL	PASS

STERILITY TEST (BATCH & SIZE COMPLIANCE)	USP 71 COMPLIANT	METHOD SUITABILITY ON FILE	Yes
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Samples accepted by Pharmetric Lab are assumed to be prepared by the customer within the guidelines of USP <71> regarding sample size and number of units sampled. Pharmetric conducts sterility testing using bioluminescence technology. Per USP <797> "A method not described in the USP may be used if verification results demonstrate that the alternative method is at least effective and reliable as the USP membrane filtration method as described in USP<71>." Sterility Testing performed by Pharmetric Lab using bioluminescence technology was validated as equal to compendial test by direct comparison. Validation documents are on file at Pharmetric Lab, LLC.

TEST DATE refers to start date. Please note some tests are carried out over multiple days.

The results on this CoA relate only to the item(s) tested and apply to the sample as received. This Report shall not be reproduced except in full without approval of the laboratory.



ACCREDITED TESTING LABORATORY. ACCREDITATION NUMBER 5386.01

COMMENTS	QUALITY SIGNATURE
	Yovany Gomez Quality Assurance Associate 1 5/5/2026 6:38:38 PM 



PHARMETRIC
LABORATORY

CERTIFICATE OF ANALYSIS (CoA)

Pharmetric Laboratory
7265 Ulmerton Road
Largo, FL, 33771

Issue #:	0	SAMPLE #:	S429007
CoA RELEASE DATE:	08 May 2026		

CLIENT:	Logos Pharmacy	CONTACT:	Carlos Martinez
CLIENT ADDRESS:	7543 W Waters Ave, Tampa, FL, 33615		
DESCRIPTION:	Semaglutide/Glycine 5mg/20mg/ml	LOT #:	LG396001001
STRENGTH:	5 mg/mL - 20 mg/mL	CONTAINER:	2 mL Vial
STORAGE CONDITIONS:	Refrigerated (2°C to 8°C)	BATCH SIZE:	250
DATE(S) RECEIVED:	30 Apr 2026	EXP.:	Not Provided
		QUANTITY RECEIVED:	12

TEST DATE	TEST	QTY	ACCEPTANCE CRITERIA	RESULTS	COMPLIANCE STATEMENT
05 May 2026	Sterility-ATP Bioluminescence Test	10	No Evidence of growth	No Evidence of growth	PASS
07 May 2026	USP <85> Endotoxin Test	1	350 EU/mL	< 25.0 EU/mL	PASS

STERILITY TEST (BATCH & SIZE COMPLIANCE)	USP 71 COMPLIANT	METHOD SUITABILITY ON FILE	Yes
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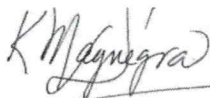
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ACCREDITED TESTING LABORATORY. ACCREDITATION NUMBER 5386.01

COMMENTS	QUALITY SIGNATURE
	Karelia Maynegra Quality Assurance Manager 5/8/2026 11:10:52 AM 



PHARMETRIC
LABORATORY

CERTIFICATE OF ANALYSIS (CoA)

Pharmetric Laboratory
7265 Ulmerton Road
Largo, FL, 33771

Issue #:	0	SAMPLE #:	S429006
CoA RELEASE DATE:	06 May 2026		

CLIENT:	Logos Pharmacy	CONTACT:	Carlos Martinez		
CLIENT ADDRESS:	7543 W Waters Ave, Tampa, FL, 33615				
DESCRIPTION:	NAD+ SOLUTION 100MG/ML (10ML) INJECTABLE	LOT #:	04162026@10		
STRENGTH:	100mg/ml	CONTAINER:	10 mL Vial		
STORAGE CONDITIONS:	Refrigerated (2°C to 8°C)	BATCH SIZE:	250	EXP.:	Not Provided
DATE(S) RECEIVED:	30 Apr 2026	QUANTITY RECEIVED:	11		

TEST DATE	TEST	QTY	ACCEPTANCE CRITERIA	RESULTS	COMPLIANCE STATEMENT
06 May 2026	Sterility-ATP Bioluminescence Test	10	No Evidence of growth	No Evidence of growth	PASS
02 May 2026	USP <85> Endotoxin Test	1	350 EU/ml	4.07 EU/mL	PASS

STERILITY TEST (BATCH & SIZE COMPLIANCE)	USP 71 COMPLIANT	METHOD SUITABILITY ON FILE	Yes
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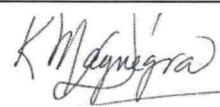
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COMMENTS	QUALITY SIGNATURE
	Karelia Maynegra Quality Assurance Manager 5/6/2026 12:49:15 PM 



PHARMETRIC
LABORATORY

CERTIFICATE OF ANALYSIS (CoA)

Pharmetric Laboratory
7265 Ulmerton Road
Largo, FL, 33771

Issue #:	0	SAMPLE #:	S429004
CoA RELEASE DATE:	05 May 2026		

CLIENT:	Logos Pharmacy	CONTACT:	Carlos Martinez		
CLIENT ADDRESS:	7543 W Waters Ave, Tampa, FL, 33615				
DESCRIPTION:	GLUTATHIONE 200MG/ML	LOT #:	Lg396000890		
STRENGTH:	-	CONTAINER:	10.0 mL Vial		
STORAGE CONDITIONS:	Refrigerated (2°C to 8°C)	BATCH SIZE:	250	EXP.:	Not Provided
DATE(S) RECEIVED:	30 Apr 2026	QUANTITY RECEIVED:	12		

TEST DATE	TEST	QTY	ACCEPTANCE CRITERIA	RESULTS	COMPLIANCE STATEMENT
05 May 2026	Sterility-ATP Bioluminescence Test	10	No Evidence of growth	No Evidence of growth	PASS
02 May 2026	USP <85> Endotoxin Test	1	35 EU/mL	5.18 EU/mL	PASS

STERILITY TEST (BATCH & SIZE COMPLIANCE)	USP 71 COMPLIANT	METHOD SUITABILITY ON FILE	Yes
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COMMENTS	QUALITY SIGNATURE
	Yovany Gomez Quality Assurance Associate 1 5/5/2026 3:39:43 PM 



PHARMETRIC
LABORATORY

CERTIFICATE OF ANALYSIS (CoA)

Pharmetric Laboratory
7265 Ulmerton Road
Largo, FL, 33771

Issue #:	1	SAMPLE #:	S423999
CoA RELEASE DATE:	22 Apr 2026		

CLIENT:	Logos Pharmacy	CONTACT:	Carlos Martinez
CLIENT ADDRESS:	7543 W Waters Ave, Tampa, FL, 33615		
DESCRIPTION:	SERMORELIN 2MG/ML (5ML VIAL)	LOT #:	04162026@12
STRENGTH:	2MG/ML	CONTAINER:	5.0 mL Vial
STORAGE CONDITIONS:	Refrigerated (2°C to 8°C)	BATCH SIZE:	250
DATE(S) RECEIVED:	16 Apr 2026	EXP.:	Not Provided
		QUANTITY RECEIVED:	12

TEST DATE	TEST	QTY	ACCEPTANCE CRITERIA	RESULTS	COMPLIANCE STATEMENT
21 Apr 2026	Sterility-ATP Bioluminescence Test	10	No Evidence of growth	No Evidence of growth	PASS
20 Apr 2026	USP <85> Endotoxin Test	1	350 EU/mL	< 2.50 EU/mL	PASS
22 Apr 2026	Potency Assay HPLC	1	Sermorelin Acetate 2mg/mL (90.0-110.0%)	Sermorelin Acetate [2.0453 mg/mL (102.3%)]	PASS

STERILITY TEST (BATCH & SIZE COMPLIANCE)	USP 71 COMPLIANT	METHOD SUITABILITY ON FILE	Yes
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RE-RELEASE EXPLANATION	Fill volume update
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Samples accepted by Pharmetric Lab are assumed to be prepared by the customer within the guidelines of USP <71> regarding sample size and number of units sampled. Pharmetric conducts sterility testing using bioluminescence technology. Per USP <797> "A method not described in the USP may be used if verification results demonstrate that the alternative method is at least effective and reliable as the USP membrane filtration method as described in USP<71>." Sterility Testing performed by Pharmetric Lab using bioluminescence technology was validated as equal to compendial test by direct comparison. Validation documents are on file at Pharmetric Lab, LLC.

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COMMENTS	QUALITY SIGNATURE
	Yovany Gomez Quality Assurance Associate 1 4/24/2026 1:06:57 PM